Vision Anomaly Detection Literature Review

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1 Introduction

Natural disasters represent some of the most destructive forces on Earth, posing significant threats to human life, infrastructure, and ecosystems globally. The devastating impact of such events has spurred growing interest in automated, intelligent monitoring systems capable of early detection and rapid situational assessment. The ability to identify precursory visual signals of disasters, such as ash plumes from volcanic eruptions, anomalous precipitation patterns, or hazardous particulate fog, is thus a critical research objective.

In order to develop a comprehensive agentic LLM-based system that can be deployed on low-powered edge devices to meet those needs, one major hurdle is the development of automated anomaly detection. These anomalies span the modalities of images and videos, where they may reveal indications of potential hazards visible from miles away such as a volcanic eruption, an onset of heavy rainfall, or dense fog caused by particulate matter. There are two ideal approaches: one involves the invocation of tool-calls from an agentic framework, while the other is an end-to-end system. This synthesis was conducted for the purposes of understanding the fundamental structure involved in constructing visual anomaly detection methods and the challenges or assumptions involved such that it may serve as a launchpad into current and future works. Anomaly detection covers a plethora of subtopics, and a comprehensive look into the entire field is intractable. Thus, to serve as a launchpad for progressing the vision-related subsection of the field, this paper shall be a survey of recent techniques and publications that have had time to mature and evolve into seminal works. This review shall also be constrained to vision tasks where the model's inputs are primarily image or video feeds. Outputs are varied and may range from binary classification to saliency maps.

Research in the past few years dealt with creating a model from an amalgamation of techniques while addressing key novel issues. However, looking forward, researchers recognize that VLMs and multi-modal LLMs tend to show promise, while other researchers aim to improve the status quo.

2 Paper Search

2.1 Common Tasks & General Knowledge & Seminal Works

In an effort to discover the common tasks for visual anomaly detection, a broad search for survey papers was conducted. For each paper, a brief inspection of the abstract was applied to determine the primary scope of the paper. Each paper must have its primary focus be in regards to only vision anomaly detection methods, targeting image/video tasks. Access restricted papers were ignored. ArXiv papers and duplicate papers were removed. The contents of the remaining papers were reviewed and referenced papers were then investigated further. The only exception was that one of the survey papers mentioned medical imaging, but lacked significant details. Therefore to maintain its representation, an exclusive search was conducted to acquire a medical survey paper. Seminal works are extracted from the overlapping references in the survey papers, typically associated with high citation count, above 400 citations on Semantic Scholar.

GOOGLE SCHOLAR For GOOGLE SCHOLAR, the following search query was used ("visual" OR "vision") ("anomaly" OR "novelty") ("detection" OR "localization") ("survey" OR "review" OR "comparative analysis") and sort by relevance, the first 10 papers were considered.

LENS For LENS.ORG, the following search query was used (title:(Vision Anomaly Detection Survey) OR abstract:(Vision Anomaly Detection Survey) OR keyword:(Vision Anomaly Detection Survey) OR field_of_study:(Vision Anomaly Detection Survey)) and sorted by relevance (based on query matching score used in Elasticsearch), the first 10 papers were considered.

IEEE For IEEE, the following search query was used ("All Metadata":"visual" OR "All Metadata":"vision") AND ("All Metadata":"anomaly" OR "All Metadata":"novelty") AND ("All Metadata":"detection" OR "All Metadata":"localization") AND ("All Metadata":"survey" OR "Title":"review" OR "Title":"comparative analysis") and sorted by relevance, the first 10 papers were considered.

Recent Works Pertaining to recent works and methodologies, not all recent papers had time to mature, therefore citations as a heuristic fail. Thus to address this shortcoming, during the search, additional papers that followed the recommended trend set by survey papers were added to improve understanding in those directions.

Use of AI AI tools were used in three capacities during the preparation of this review. First, as a semantic search aid to locate relevant passages within papers, with all retrieved content verified against the original source text. Second, as a sanity check for conceptual understanding of unfamiliar topics, with claims cross-referenced against the primary literature before inclusion. Third, for language and clarity editing of the manuscript.

3 Domains

Each paper that is derived from the seminal works involves one or more applicable domains. To be included under a given domain, a paper must either directly address it or demonstrate anticipated applicability through sufficient supporting evidence such as additional dataset evaluations, detailed observations, or a well-reasoned theoretical justification explaining how the method transfers to that domain. This bar exists because papers frequently reduce cross-domain applicability to a single speculative sentence. A passing mention of interest or motivation is not sufficient; concrete action or evidence is required. For example, a paper focused on defect detection in manufacturing clearly falls under Engineering and Technology, but a secondary claim of applicability to medical imaging is only counted if the authors substantiate it. The domains listed here do not encapsulate the entire field, but reflect those most prominently represented in the literature.

Industrial Covers the spectrum of industrial engineering and manufacturing, which is prominently featured in a majority of the seminal papers, along with robotics and autonomous systems.

Physical sciences Relates to the non-biological natural sciences such as material science, surface engineering, which in the context of anomaly detection, deals with surface defects and textiles. This section is also inclusive of astronomy, astrophysics, environmental science, and remote sensing.

Life sciences Encompasses the understanding of biological sciences such as biology and ecology.

Health sciences Differs from life sciences in that it deals with the applicability of knowledge to actualized care. Anomaly detection in this field deals with the diagnostics of medical data, medical imaging, and practical health care.

Social sciences Covers the scope of human behavioral sciences, economics, forensics, criminology, and public safety. Other interdisciplinary topics included under this field are transportation safety and traffic management.

4 Task Taxonomy

Visual anomaly detection serves many purposes ranging from satellite imagery, typically referred to as change detection, to industrial settings. The following is a list of common tasks visual anomaly detection has been used for. Solely based on survey paper count, public safety (people and roads) garnered the most attention with manufacturing coming second. Survey papers that maintain a primary focus will be considered for the task taxonomy. In the event that a paper's primary topic is too broad and coarse, it will not be included.

Table 1: Visual tasks derived from survey papers that have a domain specific focus.

Task	Domain	Description	Survey References
Industrial Manufacturing	Industrial	Quality assurance in manufacturing; detecting defects in production and manufacturing. Also used for visual inspection in routine maintenance for facilities. Automated Inspections.	[1][2][3]
Remote Sensing	Physical	Hyperspectral images, often comprised of hundreds of spectral bands, captured by satellite instruments to detect anomalies such as military operations, mineral deposits, maritime rescue, etc.	[4]
Urban	Social	Urban environments involve detection of anomalies as it relates to public safety, society, roads, and traffic. Such anomalies include property damages, violence, robberies, traffic violations, etc. Typically involves people and crowds.	[5][6][7] [8][9]
Medical Imaging	Health	These tasks are related to medical imaging such as CT and MRI scans and detecting lesions.	[10]
Inspection	Industrial	Inspections pertaining to infrastructure, specifically the survey paper focuses on contact lines.	[11]

4.1 Public Safety, People, Roads

Public safety anomalies deal with motor vehicles and city scapes. Anomalies may include traffic violations such as illegal U-turns, over-speeding, vehicles crossing stop lines or accidents [7]. Mayank Srivastava et al. referenced [6] Sultani et al. to describe that anomalies for this category also covers assault, arrest, arson, shooting, accident, burglary, explosion, shoplifting, fight, robbery, abuse, stealing, and vandalism. Furthermore, Santhosh et al.'s work describes the idea of three distinct anomaly types. Point anomalies are single isolated events like a car driving on the wrong side of the road. Contextual anomalies are anomalies conditioned on the context. To clarify the author's example, a biker traveling at a constant speed of 25 mph may be considered normal if the traffic surrounding the biker is also traveling around 25 mph. Conversely, if the traffic is at 5 mph and the biker travels at 25 mph, the biker's relative speed is significantly higher than the surrounding traffic and this may be considered an anomaly [7]. This is mirrored in Patrikar et al.'s and Zaheer et al.[5] work where they mention how 60 km/hr highway speeds are considered normal in most cases, but would be abnormal for a crowded street[12]. Lastly, collective anomaly is defined as an anomaly arising from a collective group of instances. To clarify, a stretch of road, where there is typically little traffic suddenly becomes inundated with many vehicles causing stop and go traffic. In this scenario, the anomaly is formed by the surge of individual vehicles piling up [7].

Detecting anomalies for public safety and surveillance is motivated by the impracticalities of manual monitoring on a massive scale [6][5]. However, through Srivastava et al.'s experiments for this task, robust detection still remains a challenge as the authors only report an accuracy of about 43% [6]. Duong et al. highlight some challenges as realtime constraints, time sensitive, high power consumption, feature extraction due to occlusion, overlap, clutter, noise, and classification ambiguity for intraclass and interclass similarity. Intraclass variation refers to differences within a class — for example, red and green apples are still apples. Conversely, Interclass is where some apples may appear identical to oranges in certain conditions. Essentially, the model gets confused where to draw the line for variations and objects bleeding into another class. [8].

Prior to 2019, most approaches addressed this problem by modeling a normal distribution from training data and flagging deviations in test samples as anomalous events [5]. Representative architectures explored in the literature include CNNs, CNN-LSTM hybrids, Conv3D-LSTM networks, autoencoder-SVM pipelines, ConvLSTM autoencoders, stacked RNNs, and GANs [5]. As Patrikar et al. note, methods leveraging CNNs, LSTMs, GANs, and other deep learning architectures show strong promise, achieving superior accuracy across most surveillance scenarios [12].

4.2 Manufacturing

As noted by Cui et al., anomaly detection offers significant practical benefits in manufacturing and production by reducing reliance on human decision-making and specialized technical expertise, thereby lowering labor costs. By adopting AI-based systems, manufacturers can also improve consistency while reducing variance and subjective judgment between human operators. Safety considerations further drive the field of anomaly detection, particularly in maintaining strong focus on identifying material anomalies. Another factor that distinguishes manufacturing from many other domains is the unusual abundance of defect imagery [2].

Tao et al. and Li et al. describe two types of anomalies in this domain: textural and functional, both defined by their deviation from normal samples[1][2]. Textural anomalies contain low amounts of semantic information and typically manifest as small cracks, holes, or scratches. Functional anomalies carry a higher degree of semantic information, such as missing, misplaced, or misaligned components[1][3].

Cui et al. discussed some challenges being balancing the false positive alarm rate and missed alarm rate, where missed alarms lead to commercial loss from inferior products, but a high alarm rate will lead to increased costs in manual confirmations. Scarcity of labeled defect data and existing benchmarks are relatively simple and do not accurately reflect the complexity of actual production (also mentioned by Zhuo Li et al.[3]). Real world tasks may involve detecting impurities in edible oil, internal engine lining defects, wine impurities, bearing defects. Furthermore, difficulties may include small defect sizes, object appearance variability, and texture differences across materials[2]. Xian Tao et al. also mention another set of challenges which involves shifting the model's latent space to be compatible with human experience. For normal data, improper variations in illumination, perspective, scale, shadows, blur, etc. can result in significant differences that may cause false positives. Second involves a multi-scale problem where a model needs to be able to handle defects spanning a few pixels to larger defects that cover the majority of the image. Lastly, scarcity of pixel-level labeled data makes precise segmentation of anomalous contours unlikely. Therefore the decision boundary will struggle to equal the ground truth distribution boundary[1].

Cui et al. surveyed how normalizing-flow-based methods are used to convert a simple distribution into a complex distribution by applying a series of invertible transformations such that distribution of normal samples is learned, conversely normalizing flows can also be used to learn the distribution of artificially augmented defects. Utilizing both is denoted as joint learning and improves the generalization ability of the model. Secondly, Representation-based methods apply a pre-trained model to extract image features for anomaly detection. These pretrained, or foundation models garner massive amounts of data which allows them to capture a broad range of patterns and relationships. Enhanced by finetuning the model on specific tasks, may allow the model to quickly adapt to new domains and produce high-quality representations[2]. Lastly, Cui et al. mentioned multi-modal learning as a method to facilitate deeper understanding with the introduction of different modalities such as image and text[2]. Representative models spanning these paradigms include VAEs, GANs, PatchCore, PaDIM, SPADE, DRAEM, and CutPaste [2].

4.3 Hyperspectral

Hyperspectral images, are often comprised of hundreds of spectral bands, captured by remote sensing instruments. These spectral bands are similar to color channels in a traditional RGB image, but are not limited to visible light. Hyperspectral image detection is classified into spectral matching for known targets and anomaly detection for unknown targets. Anomalies are determined by low probability deviations from the hyperspectral background distribution. These deviations may manifest as military targets, mineral exploration, maritime rescue, etc. Hao Fan et al. describe that hyperspectral data contains two types of information: spatial to determine position, and spectral which contains reflective characteristics [4].

The authors posed, in their survey of hyperspectral anomaly detection, three methods to address hyperspectral imaging anomalies: statistical methods, representation methods and deep learning methods. Statistical, defined by the authors, follows a specific Gaussian probability distribution where anomalies deviate from it. The logic is simple, but often fails to properly map complex background distributions. Representation based methods, utilize approximation by linear combination from known background elements in the dictionary. If the pixel can be approximated well it is considered normal, otherwise it is anomalous. Representation is more adaptable, but requires solving complex optimization problems, resulting in high computational complexity and insufficient real-time performance, with additional cost of manual parameter tuning. Lastly, deep learning methods, being the most recent, utilizes feature learning and learn the background from the data. Deep learning methods are capable in learning complex patterns but require a significant computation and data resources [4].

4.4 Infrastructure Inspection - Contact Lines

Overhead contact lines (OCLs) are electrified wires suspended above railways or roadways to supply vehicles with electric power. Power is transmitted through the contact between the vehicle's deployed pantograph and the contact wire. Because OCLs are continuously exposed to dynamic pantograph excitations, extreme weather, and environmental stressors, they are prone to localized anomalies that pose latent risks of failure if left unaddressed, even if there is no immediate power transmission disruptions [11]. In this context, an anomaly refers to either a deviation of a measured parameter—geometrical (e.g., contact wire height, stagger, residual wear), electrical (e.g., arcing between the wire and pantograph strip, hot spots), or mechanical (e.g., contact force, pantograph head acceleration)—from its nominal range, or a defective condition of an OCL component such as cracks, looseness, abrasions, missing parts, deformations, or shifts in cross-span and contact line equipment [11].

Yu et al. describe the unique challenges of this class of anomaly detection task. Firstly, cameras mounted on high-speed trains are affected by sway and vibrations causing geometric measurement errors, blurry imagery, and compensation algorithms only factor in tilt but not pitch and yaw. Secondly, from the author's surveyed papers, imagery is in mostly similar positions and lighting which may not reflect real world environment variations. Real world cases suffer from the influences of weather, illumination, and complex backgrounds. Thirdly is the nature of anomaly detection, where some defects are extremely rare such as a crack on a swivel only occurring once on the Guangzhou railway during the last three years. Fourthly is the real time computation constraint, where the model is expected to output a response within a set time frame due to the travel speed of trains ranging from 120km/h to 350km/h. Fifth, perspective shifts can cause false positives, where from certain angles, any slight disengagement of contact points may visually appear still in contact. Lastly is the lack of datasets present for OCL tasks [11].

Yu et al. categorize computer vision approaches for OCL anomaly detection into stereo vision-based methods, which reconstruct 3D geometry via triangulation to measure parameter deviations, and object vision-based methods, which apply cascaded CNN pipelines—typically Faster R-CNN—for component detection and defect identification. To mitigate the scarcity of labeled defects, they advocate semi-supervised and unsupervised strategies such as autoencoders and GAN-based augmentation [11].

4.5 Medical

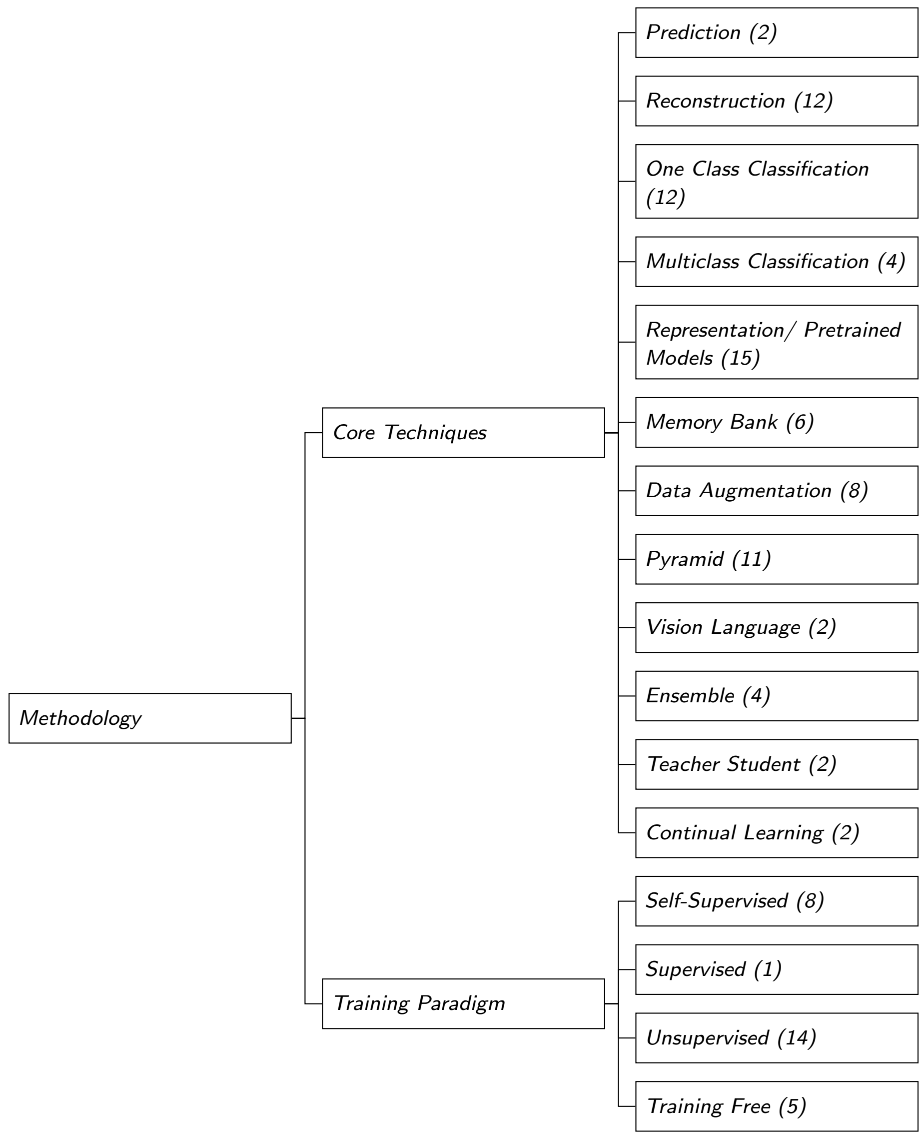
According to Cai et al., there are two types of anomalies as it relates to the medical field. Local anomalies and global semantic anomalies. Local anomalies are usually small lesions within healthy tissue. Global semantic anomalies are abnormalities that cover the entire image. [10].

Cai et al. identify several open problems in medical anomaly detection. Many widely used benchmarks such as Hyper-Kvasir and OCT2017 are "remarkably easy for anomaly detection", undermining their value for evaluation. The core reconstruction assumption is also fragile: a model trained on normal images may either generalize too aggressively and faithfully reconstruct anomalies (false negatives), or accumulate intrinsic reconstruction error on normal regions that masks true lesions (false positives). Latent space configuration remains dataset-dependent, with no principled way to select an optimal compression rate without test-time tuning, and the choice of distance function for measuring reconstruction error is similarly inconsistent across modalities, with no single metric dominating. Frozen ImageNet-pretrained encoders prove surprisingly effective—often outperforming bespoke self-supervised methods but the path to improving on this baseline is unclear. Denoising diffusion probabilistic models (DDPMs) are constrained by their noise schedule, working only when the injected noise is sufficient to corrupt the anomalous region prior to reconstruction. Notably, simple intensity thresholding achieves the second-best segmentation performance on BraTS2021, outperforming more sophisticated DDPM-based pipelines. Finally, volumetric scans are typically reduced to 2D axial slices, discarding information along the third dimension [10].

Cai et al. benchmark thirty representative methods across seven medical datasets spanning five imaging modalities, finding that no single approach dominates across all settings. In the absence of pretraining, methods that learn to regenerate normal data prove more robust than those trained to discriminate synthetic pseudo-anomalies, and the simple autoencoder remains a strong baseline. When synthetic anomalies are used, realism of the synthesis and a two-stage training procedure (representation learning followed by a one-class classifier) outperform direct one-stage discrimination. Most strikingly, leveraging ImageNet-pretrained weights—whether for distance measurement, input transformation, or direct feature extraction—yields the strongest overall results, suggesting that the discriminative power of pretrained features, rather than any particular detection paradigm, is the dominant factor in current medical anomaly detection performance [10].

5 Methodology and Approaches

When considering model architecture, there are a plethora of approaches and aspects to be cognizant of. Typically requirements will guide the research to choose a selection of techniques for the development of a pipeline then apply a novel improvement to any given section of the pipeline.



Method	Techniques	Training	Domain
Future Frame Prediction [13]	Prediction Representation	Self-Supervised	Social
STAE [14]	Prediction Reconstruction	Unsupervised	Social
DRÆM [15]	Reconstruction Data Augmentation	Self-Supervised	Engineering, Physical

Method	Techniques	Training	Domain
Reverse Distillation [16]	Reconstruction One Class (Distribution) Representation Pyramid Teacher Student	Unsupervised	Engineering
Stacked RNN [17]	Reconstruction Representation Pyramid	Unsupervised	Social
MemAE [18]	Reconstruction Memory Bank	Unsupervised	Engineering (Cybersecurity), Social
GANomaly [19]	Reconstruction One Class (Distribution)	Unsupervised	Social
AE (SSIM) [20]	Reconstruction	Unsupervised	Industrial, Physical
AnoGAN [21]	Reconstruction One Class (Distribution)	Unsupervised	Health
Spatiotemporal Autoencoder [22]	Reconstruction	Unsupervised	Social
2D CNN AE [23]	Reconstruction	Unsupervised	Social
RIAD [24]	Reconstruction Data Augmentation Pyramid Ensemble	Self-Supervised	Industrial, Social
*MaskVIT [25]	Reconstruction Multiclass Representation Pyramid	Unsupervised	Industrial
PatchCore [26]	One Class (Distribution) Representation Memory Bank Pyramid	Training Free	Engineering, Social
SPADE [27]	One Class (Distribution) Representation Memory Bank Pyramid	Training Free	Engineering, Life, Social
PaDiM [28]	One Class (Distribution) Representation Pyramid	Training Free	Engineering, Social
CutPaste [29]	One Class (Distribution) Representation Data Augmentation Ensemble	Self-Supervised	Engineering

Method	Techniques	Training	Domain
FastFlow [30]	One Class (Distribution) Representation Pyramid	Unsupervised	Engineering
CD-SVDD [31]	One Class (Binary)	Unsupervised	Industrial, Physical, Life, Health, Social
GEOM [32]	One Class (Distribution) Multiclass Data Augmentation	Self-Supervised	Social
PANDA [33]	One Class (Binary) Representation Continual Learning	Unsupervised	Industrial, Physical, Life, Health
CSI [34]	One Class (Binary) Data Augmentation Ensemble	Self-Supervised	Industrial, Health, Social
*AnomalyGPT [35]	Multiclass Representation, Memory Bank Data Augmentation Pyramid Vision Language	Self-Supervised	Industrial
*IAD-GPT [36]	Multiclass Representation Memory Bank Data Augmentation Pyramid Vision Language	Self-Supervised	Industrial
AnomalyDino [37]	Representation Memory Bank Data Augmentation	Training Free	Engineering
Uninformed Students [38]	Representation Pyramid Ensemble Teacher Student	Unsupervised	Engineering
MIL [39]	Representation	Supervised	Social
*PatchCoreCL [40]	Continual Learning (and more)	Training Free	Industrial, Health

5.1 Core Architecture

Each subsection presents a core strategy for anomaly detection. These strategies are best understood as conceptual building blocks that can be composed into pipelines or hybrid systems rather than as mutually exclusive alternatives. For instance, a reconstruction-based approach can be combined with a pyramid method, with each component addressing a distinct limitation of the other: the pyramid supplies multi-scale spatial coverage, while reconstruction provides the anomaly signal.

5.1.1 Prediction

The prediction method aims to determine abnormal events by comparing the actual data to expected data. The discrepancy between actual and expected can serve as the anomaly signal. This method is used with sequential data such as a video stream.

Future Frame Prediction[13] Given consecutive sequential frames (I_1,I_2,...I_t), the model learns to predict the next frame (I_{t+1}). If the prediction closely matches the actual frame, the event is most likely normal. If there is a large discrepancy, it is marked as abnormal.

STAE[14] Prediction decoder forecasts future frames using a weight-decreasing loss that assigns progressively lower importance to distant frames, focusing the model on learning motion trajectories of existing objects rather than the unpredictable appearance of new objects entering the scene.

5.1.2 Reconstruction

The underlying philosophy of reconstruction-based methods is that an image can be decomposed into an intermediary representation that is trained exclusively on known normal patterns. At inference time, the reconstructed image would have its anomalies suppressed since the image was compressed into an intermediate representation that can only represent normal patterns. Comparing the difference between the input and output images/features produces a reconstruction error that may serve as an anomaly signal.

PCA and VAE methods leverage a bottleneck to compress the input into a lower dimensional representation. Since the bottleneck is trained exclusively on normal data, it learns to only preserve the most prominent normal structures. Anomalous features, lacking representation in this learned subspace, are effectively discarded during compression. The reconstructed image is therefore constrained towards normality, and the anomaly score is the difference between the input and reconstructed output. Diffusion relies on introducing noise to obfuscate and corrupt the input. If the anomaly is sufficiently obfuscated by the noise, the denoising process will reconstruct the image as normal since the model only learned to denoise normal patterns. The anomaly score then is the difference between the input and the reconstructed output.

DRÆM[15] Reconstructive sub-network, U-Net is utilized to restore anomaly free patterns.

Reverse Distillation [16] Though the authors distinguish their method from traditional reconstruction approaches, the student decoder is fundamentally reconstructing the teacher's feature representations from a compressed bottleneck embedding. The anomaly signal comes from the failure to reconstruct anomalous features.

Stacked RNN[17] Contains a learned dictionary of normal data (video frames) with size 2048x2048 serving as sparse representation or bottleneck. Anomaly signal is the reconstruction error between the input feature and reconstructed dictionary.

MemAE[18] Utilizes a memory-augmented autoencoder where encodings query a memory bank of prototypical normal patterns via sparse attention-based addressing, constraining reconstruction toward normality and amplifying reconstruction error on anomalies.

GANomaly[19] Bottleneck autoencoder reconstructs input; anomaly from reconstruction difference.

AE (SSIM)[20] Based on autoencoder to reconstruct, anomaly signal derived via SSIM (structural similarity index) to compare luminance, contrast, and structure, replacing the typical per-pixel loss.

AnoGAN[21] AnoGAN reconstructs query images via the generator and uses the residual image as the anomaly signal, the difference between input and its normal reconstruction.

Spatiotemporal Autoencoder[22] Uses a spatial encoder to learn and encode the feature space then applies a ten frame temporal encoder decoder to capture how spatial features in frames evolve with time. Then these ten reconstructed frames are compared with the input to acquire the anomaly signal

2D CNN AE[23] Uses a histogram of orientated gradients (HOG) and histogram of optical flow (HOF) trajectory features and a fully connected convolutional autoencoder on raw temporal cuboids of T stacked frames. HOG+HOF reconstructs the features for comparison while the Conv Autoencoder alternative method reconstructs the image for comparison.

STAE[14] Uses a 3D convolutional autoencoder to reconstruct input video clips while preserving temporal information across all layers by convolving in the time dimension. Which allows for reconstruction error to capture motion based anomalies.

RIAD[24] RIAD replaces standard autoencoder reconstruction with blind inpainting, where masked pixels are never visible to the network, preventing the over-faithful anomaly reconstruction that plagues conventional autoencoders.

***MaskVIT[25]** MaskViT injects stage-specific mask matrices directly into the decoder's self-attention (neighbor masks for deep semantic features, random masks for shallow texture features) to prevent the identical shortcut problem where reconstruction networks trivially copy input to output.

5.1.3 One Class Classification: Binary and Distribution

One-class classification methods involve learning a representation of a single normal homogeneous class during training, such that classifications falling outside the normal class decision boundary or in low-density regions of the distribution are flagged as anomalous. The core difference between the decision boundary and distribution is that one is for binary classification, while distributions are probabilistic, pulling from density estimation.

PatchCore[26] Distribution based on class outlier detection in feature space.

SPADE[27] KNN distance as density based anomaly score.

PaDiM[28] Decomposes images into a spatial grid of local regions and fits a multivariate Gaussian per patch position, modeling normality independently at each location. Anomaly scores are computed via Mahalanobis distance, comparing each test patch against its position-specific learned reference to produce location-sensitive anomaly maps.

CutPaste[29] Distribution, learned representation that uses Gaussian Density Estimator on features/representations, scores anomalies by log-likelihood.

FastFlow[30] Distribution, uses normalizing flows to model the probability density of normal features then uses likelihood as the anomaly score.

Reverse Distillation [16] The One-Class Bottleneck Embedding (OCBE) module projects features into a compact one-class embedding space trained exclusively on normal data.

CD-SVDD[31] A one-class binary classification method that maps input data through a neural network into a lower dimensional feature space and learns a minimum-volume hypersphere to enclose normal data, classifying samples outside the boundary as anomalous.

GANomaly[19] Rather than comparing input and reconstructed images at the pixel level, a second encoder re-encodes the reconstructed image back into latent space, and the anomaly score is the distance between the two latent vectors.

AnoGAN[21] Distribution, Trained exclusively on healthy data; the anomaly score measures how well a query fits the learned normal distribution. The discrimination loss checks whether the generated image lies on the learned manifold.

GEOM[32] Test images are scored by how likely their softmax patterns are under Dirichlet distributions fitted to normal data, flagging low-likelihood samples as anomalous.

PANDA[33] One class binary. PANDA minimizes the loss to pull all normal training features toward their mean center c, so anomalous test samples land further from c and are detected via kNN distance.

CSI[34] One class binary, CSI learns a contrastive embedding of normality and flags test samples that fall far from it, replacing shallow objectives like DeepSVDD's hypersphere with shift-aware instance discrimination.

5.1.4 Multiclass Classification

Multiclass classification methods frame anomaly detection as the task of learning discriminative representations across multiple distinct object or texture categories simultaneously, rather than training a separate model per class. A single unified model is trained to classify inputs into one of several known normal categories, and anomaly scoring is derived from classification uncertainty or feature-space distances. The key advantage is scalability: a single model can serve as the anomaly detector for many product types and can output the type of anomaly.

GEOM[32] GEOM distinguishes among 72 transformation indices forces the network to learn class-specific geometric features, which then serve as the anomaly signal through their softmax statistics.

***AnomalyGPT[35]** Trains one model on all 15 MVTec-AD categories simultaneously, replacing per-class models with a single LLM that directly outputs binary anomaly judgments.

***IAD-GPT[36]** Trains one model on all categories simultaneously, using APG's per-class prompts to maintain category-specific sensitivity.

***MaskVIT[25]** MaskViT trains a single unified model across all product categories simultaneously (15 for MVTec-AD, 12 for VisA), achieving 99.1% image-level AUROC on MVTec-AD without per-class fine-tuning.

5.1.5 Representation/Pretrained Models

This involves the utilization of a pretrained network which has been trained on a large dataset such as ImageNet and has developed a rich hierarchical feature representations of the related task. By applying transfer learning, the weights of the pretrained backbone are frozen, thereby preserving the learned representations, upon which additional components such as adapters, projection heads, and task-specific layers can be built and trained. This allows the model to be adapted to downstream tasks while preventing catastrophic forgetting. Alternatively, fine-tuning unfreezes some or all layers of the pretrained backbone, allowing the weights to be updated during training. This enables the model to further adapt its representation to the target domain while leveraging the rich features learned during pretraining.

AnomalyDino[37] Utilizes DinoV2 (foundational model) as a backbone, frozen weights.

PatchCore[26] Pretrained frozen backbone imagenet (WideResNet-50).

SPADE[27] Wide-ResNet50x2 pretrained on ImageNet with frozen weights as feature extractor

PaDiM[28] Frozen pretrained CNN for embedding extraction.

CutPaste[29] Pretrained EfficientNet-B4 (ImageNet) as frozen or finetuned backbone.

FastFlow[30] Frozen ImageNet, fast flow is built on top as a task-specific component (backbone + adapter pattern).

Reverse Distillation[16] Frozen WideResNet50 or ResNet trained on ImageNet.

Stacked RNN[17] Pretrained CNN as frozen feature extractor trained on the UCF101 dataset with output features of 7x7x2048.

Uninformed Students[38] Teacher is built by distilling a frozen pretrained ResNet-18 (ImageNet).

MIL[39] Frozen C3D backbone with task-specific fully connected layers on top.

Future Frame Prediction[13] FlowNet used a pretrained frozen network.

PANDA[33] PANDA fine-tunes an ImageNet-pretrained ResNet (layers 3 and 4 only) on the target normal data using a compactness loss, rather than training features from scratch with self-supervised tasks.

***AnomalyGPT[35]** Freezes ImageBind-Huge and Vicuna-7B entirely; only the decoder and prompt learner (lightweight additions) are trained.

***IAD-GPT [36]** Freezes ImageBind-Huge as image encoder and uses CLIP for text-image alignment, building TGE, decoder, and MMF as trainable layers on top.

***MaskVIT [25]** MaskViT freezes a 12-layer DINO-pretrained ViT as its encoder, leveraging its self-supervised visual representations without fine-tuning, and builds trainable fusion and decoding components on top.

5.1.6 Memory Bank

Rather than encoding normality implicitly into the model weights during training, normal patterns are stored in an explicit, structured data structure that is separate from the core network parameters. This memory serves as a reference bank of normal features against which test samples are compared. The memory may be populated at inference time or during training. Anomalies are identified as samples that deviate significantly from the stored representations, with the anomaly signal derived from the relationship between the test sample and the memory contents rather than from the model's internal transformations.

AnomalyDino[37] Inference time, nominal patch features are in a memory bank and anomalies are identified by comparing test patches against the memory bank's stored reference via nearest-neighbor distance.

PatchCore[26] Inference time, memory bank of nominal patch-level features against test samples compared via nearest neighbor distance. They introduced greedy Coreset memory subsampling to reduce memory bank by up to 100x while retaining nearly identical performance. Coreset subsampling is a method for selecting a small representative subset from a large set of points. The idea is to maximizes coverage of the feature space by iteratively picking points that are farthest from any already-selected point.

SPADE[27] At inference, all normal training image features are stored and the test sample is compared against them via nearest neighbor distance. They use correspondence/alignment to compare against normal representations via kNN at the pixel level.

MemAE[18] During Training, Between the encoder and decoder, MemAE implemented a memory module M to retrieve the most relevant memory via an attention-based addressing operator. The data structure (NxC matrix) stores prototypical normal patterns.

***AnomalyGPT[35]** In few-shot mode, stores patch features from normal reference images across 4 encoder stages and localizes anomalies by nearest-neighbor distance.

***IAD-GPT[36]** In the few-shot setting, stores patch features from normal samples across four encoder layers and localizes anomalies via nearest-neighbor distance.

5.1.7 Data Augmentation

Anomaly detection can be augmented by synthetically generating pseudo-anomalies during training. Since real anomalous samples are scarce or unavailable, artificial anomalies are synthesized by applying perturbations to normal images, such as cutting and pasting random patches, applying noise, or blending textures from external sources. The model is then trained to discriminate between normal and synthetically anomalous samples, learning a decision boundary that generalizes to real anomalies at inference time. Simple rotations, flips, shifts without much purpose do not count.

AnomalyDino[37] The paper explicitly applies rotational augmentations to the reference samples to synthetically diversify the memory bank's representation of normality. The intent — expanding coverage of normal appearance without additional real data — matches the taxonomy's description of augmentation to supplement scarce training signal.

DRÆM[15] Synthetic anomalies generated via Perlin noise masks blended with textures from the Describable Texture Dataset.

CutPaste[29] Pseudo anomaly generation strategy that cuts an image patch and pastes it at a random location of a large image.

GEOM[32] The entire method is built on 72 geometric transformations that generate a self-labeled classification task, making augmentation the core mechanism.

RIAD[24] RIAD generates its own training signal by randomly zeroing out grid-aligned regions of normal images, creating a self-supervised inpainting objective without needing any external anomaly samples or textures.

CSI [34] CSI synthesizes pseudo-anomalies by applying transformations that maximally shift the input distribution, then trains the model to discriminate normal from shifted samples.

***AnomalyGPT [35]** Generates synthetic defects via NSA (Cut-paste + Poisson editing) paired with textual Q&A descriptions to create training data.

***IAD-GPT [36]** Trains on synthetic anomalies generated by NSA, which uses Poisson image editing to blend defect patches more realistically than CutPaste.

5.1.8 Pyramid

Pyramid based methods leverage multi-scale feature representations to capture anomalies that manifest at various spatial resolutions. The core concept behind each layer in the pyramid is such that earlier layers capture fine-grained details such as lines, foundational geometry, local textures which become progressively more semantic and categorical deeper into the network. By jointly evaluating multiple layers of the pyramid, textural anomalies, such as localized defects, scratches and cracks are captured alongside functional anomalies such as missing components.

PatchCore[26] Layers 2 and layers 3 to capture anomalies at different scales and resolution, multiple feature hierarchies.

SPADE[27] Concatenates features from different ResNet blocks at different spatial resolutions to capture anomalies at multiple scales.

PaDiM[28] Multi-layer feature concatenation across semantic levels with inter-layer correlation modeling.

FastFlow[30] ResNet, features from three different blocks at different spatial resolutions are independently processed then averaged.

Reverse Distillation[16] Uses multi-scale feature representation from different layers of the network to capture anomalies at various resolution to generate anomaly maps from it.

Stacked RNN[17] Multiscale feature extraction: "Gradually partitioned the feature map into increasingly finer regions 1x1, 2x2, 4x4"

Uninformed Students[38] Multiple student–teacher ensemble pairs are trained with varying receptive field sizes. Each pair captures anomalies at a single scale determined by its receptive field, and the normalized anomaly scores are averaged across pairs to detect anomalies at multiple scales.

RIAD[24] RIAD ensembles across multiple masking granularities (k 2,4,8,16) and computes its MSGMS anomaly score over a 4-level image pyramid, capturing defects ranging from fine scratches to large missing components.

***AnomalyGPT[35]** Extracts and aggregates patch features from layers 8, 16, 24, and 32 of ImageBind to capture anomalies at multiple spatial scales.

***IAD-GPT[36]** Extracts features from layers 8, 16, 24, 32 and generates a mask at each scale, fused via MMF into a single embedding for the LLM.

***MaskVIT[25]** MaskViT extracts four hierarchical feature levels (F1–F4) from the ViT encoder and fuses them through the MSCAF module — a channel-wise squeeze-and-excitation mechanism — to preserve both fine-grained texture and high-level semantic information in a single compact representation.

5.1.9 Vision Language

Vision-Language methods are constructed from multi-modal models such as CLIP that jointly embed images and text into a shared latent space for semantic representation, enabling anomaly detection via natural language. Rather than training on task-specific imagery, these methods leverage the model's broad semantic understanding acquired during large-scale vision-language pretraining, using text prompts to describe normal and anomalous conditions. Due to the model's inherent fundamental understanding and generalization capabilities derived from the combination of language and vision, this paradigm enables zero-shot and few-shot anomaly detection, where zero-shot refers to inference without any examples from the target domain; thereby foregoing or reducing reliance on domain-specific imagery. At inference time, image regions whose features align more closely with anomalous text descriptions rather than with normal descriptions are flagged as anomalous. Furthermore, outputs may produce textual explanations with reasoning to improve perceived interpretability.

***AnomalyGPT[35]** AnomalyGPT embeds both images and normal/abnormal text descriptors into a shared space via the decoder's feature-text matching, and leverages the LLM to produce natural language anomaly judgments, enabling few-shot transfer to unseen object categories through in-context learning with text prompts rather than retraining.

***IAD-GPT[36]** Uses GPT to generate category-specific anomaly text prompts (e.g., cracks, tears for leather) that activate CLIP's segmentation, replacing generic normal/abnormal templates.

5.1.10 Ensemble

Ensemble methods aggregate anomaly scores from multiple independently trained models to produce a more robust and stable detection signal. Rather than relying on a single model's output, several models, potentially trained with different random seeds, augmentation strategies, or architectural configurations, each produce an anomaly score that is combined through averaging, voting, or other aggregation functions. The resulting score benefits from variance reduction across individual models, smoothing out per-model sensitivity to initialization or data sampling. Ensembling is agnostic to the underlying methodology; any combination of reconstruction, one-class classification, or augmentation-based approaches may be aggregated, making it a general-purpose scoring strategy that operates downstream of the core anomaly modeling pipeline.

CutPaste[29] CutPaste ensembles anomaly scores from five independently trained 3-way classification models, each using a different random seed, improving image-level detection from 95.2 to 96.1 AUC.

Uninformed Students[38] Uses an ensemble of M student networks to derive the anomaly signal from their disagreement

RIAD[24] RIAD averages anomaly maps produced from reconstructions at each region size k, functioning as an internal ensemble over scale that smooths sensitivity to any single masking granularity without training separate models.

CSI[34] CSI averages scores across augmented views and combines two complementary signals — contrastive similarity and shift-classifier confidence — for a more stable detection score.

5.1.11 Teacher Student

Teacher-student methods exploit the knowledge distillation framework for anomaly detection. A pretrained teacher network, which encodes rich feature representations, is paired with a student network that is trained to mimic the teacher's output on normal data. Because the student is only exposed to normal samples during training, it learns to reproduce the teacher's representations for normal inputs but fails to do so for anomalous inputs, which lie outside the student's learned mapping. The discrepancy between the teacher's and student's feature representations at inference time serves as the anomaly signal.

Reverse Distillation[16] Pretrained teacher encoder is paired with a student decoder trained to mimic the teacher on normal data. The discrepancy between their representations serves as the anomaly signal.

Uninformed Students[38] "Student networks are trained to regress the output of a descriptive teacher network... Anomalies are detected when the outputs of the student networks differ from that of the teacher network."

5.1.12 Continual Learning

Continual learning, also referred to as incremental or lifelong learning, addresses the scenario where the anomaly detection model must adapt to new object categories, domains, or defect types over time without forgetting previously learned normal representations. In industrial settings, new product lines are introduced continuously, and retraining from scratch on all accumulated data is computationally prohibitive. Continual anomaly detection methods aim to incrementally update the model as new classes arrive while preserving detection performance on previously seen categories.

PANDA[33] PANDA-EWC computes the diagonal Fisher information matrix from the ImageNet pretraining task and adds a weighted penalty to the compactness loss, slowing weight drift on parameters most important to the pretrained representation.

***PatchCoreCL[40]** Keeps total memory constant by equally rebalancing all per-task banks via coreset subsampling each time a new task arrives, since the frozen backbone makes weight-based CL techniques inapplicable.

5.2 Training Paradigm

This section covers the training paradigms used to train the architecture at a high level. The common paradigms involved in vision anomaly detection involves unsupervised, self-supervised, and semi-supervised. Reinforcement learning remains unutilized for this domain.

5.2.1 Unsupervised

Unsupervised methods operate under no availability of labeled data. The method learns patterns exclusively from unlabeled normal data, typically by modeling the distribution of normal images. During inference, samples that deviate significantly from the learned distribution are flagged as anomalous. This approach is especially valuable when anomalies are rare and unknown in advance, making it impractical to collect labeled examples.

FastFlow[30] FastFlow is trained unsupervised by maximizing the log-likelihood of normal image features under a 2D normalizing flow, learning to map the feature distribution to a standard normal without any labels or synthetic supervision signal.

Reverse Distillation[16] The student decoder and one-class bottleneck embedding are trained unsupervised by minimizing cosine similarity against a frozen pretrained teacher's multi-scale feature representations using only anomaly-free images.

Stacked RNN[17] The sRNN is trained unsupervised by jointly optimizing the dictionary and all network parameters end-to-end via reconstruction loss over unlabeled normal video frames, replacing the traditional alternating optimization of sparse coding with simultaneous SGD-based parameter learning.

CD-SVDD[31] CD-SVDD is trained unsupervised using only normal samples, where the neural network and hypersphere parameters are jointly optimized through an alternating strategy that solves the hypersphere exactly via convex dual optimization before updating the network via backpropagation.

MemAE[18] The memory contents, encoder, and decoder are jointly trained end-to-end via backpropagation on exclusively normal data, with the memory updated only through gradients from non-zero addressing weights.

GANomaly[19] GANomaly is trained in an unsupervised manner on exclusively normal data, where an adversarial autoencoder jointly optimizes reconstruction fidelity and latent space consistency without any labeled or synthetically generated anomalies.

Uninformed Students[38] An ensemble of student networks is trained unsupervised to regress the frozen teacher's feature descriptors on anomaly-free images, using the teacher's deterministic outputs as surrogate regression targets rather than any human-provided or synthetically generated labels. The remaining ensemble is trained self-supervised.

AE (SSIM)[20] Trained unsupervised on 10,000 defect-free 128x128 patches for 200 epochs using ADAM, with the autoencoder learning to reconstruct normal patterns exclusively through SSIM-based loss.

AnoGAN[21] No anomaly labels used in training. Only healthy images seen. Annotations used solely for evaluation.

Spatiotemporal Autoencoder[22] The model is trained unsupervised by minimizing the reconstruction error of a spatiotemporal autoencoder on unlabeled video volumes containing exclusively normal scenes, requiring no annotations or synthetic supervision signals.

2D CNN AE[23] Trained unsupervised using only normal video across multiple datasets simultaneously with a reconstruction objective, assuming all training events represent regular patterns without any anomaly labels or dataset-specific fine-tuning.

STAE[14] STAE is trained in an unsupervised paradigm exclusively on normal video clips, where both the reconstruction and prediction objectives derive their targets entirely from the unlabeled data itself, requiring no annotations, pretrained components, or exposure to anomalous examples.

PANDA[33] PANDA adapts pretrained features in an unsupervised manner by fine-tuning exclusively on unlabeled normal images using a compactness loss that pulls all features toward their mean, with EWC regularization requiring no target-domain labels.

MaskVIT[25] MaskViT trains its decoder and MSCAF module for 500 epochs using only cosine similarity loss between frozen encoder and trainable decoder features on defect-free images, with no labels, synthetic anomalies, or pretext objectives — learning normality purely through feature-space reconstruction error.

5.2.2 Self-Supervised

Self-supervised isn't conventionally acknowledged as a distinct category and is often classified as an unsupervised technique in the anomaly detection field [41][42]. Such is the case of DRÆM [15] which the authors classify as unsupervised. Self-supervised methods generate their own training signal by creating artificial tasks from unlabeled normal data. For example, applying known alterations to an image and training the model to identify or reverse those alterations. This exposes the model to both the synthetic data alongside the generated labels.

DRÆM[15] DRÆM is self-supervised, generating its own pixel-level supervision by synthetically corrupting normal images with randomized Perlin noise masks and external textures, then training jointly to reconstruct normality and segment the synthetic deviations.

CutPaste[29] CutPaste is trained self-supervised by formulating a proxy classification task in which a CNN learns to discriminate between original normal images and their CutPaste-augmented counterparts, with class labels generated entirely by the augmentation process itself.

Future Frame Prediction[13] The model is trained self-supervisedly on a temporal pretext task in which consecutive normal frames serve as input and the naturally occurring next frame serves as the self-derived ground truth target, with the supervision signal emerging from the temporal structure of the video itself rather than from any external annotations.

GEOM[32] The model is self-supervised, training on a pretext task of classifying which of 72 geometric transformations was applied to each normal image, using transformation indices as automatically generated labels.

CSI[34] CSI is self-supervised, generating proxy labels by applying distributional shifts (rotations) to normal images and training the encoder to both contrast and classify those shifts without any human annotation.

RIAD[24] RIAD is self-supervised because it manufactures its own training objective by randomly masking grid-aligned regions of normal images and using the original pixel values as implicit ground truth for inpainting, requiring no external labels or synthetic anomaly generation.

IAD-GPT [36] Trains across three staged phases on exclusively normal data augmented with NSA-generated synthetic anomalies and their auto-generated masks, using discriminative losses (cross-entropy, focal, dice) against self-produced labels.

AnomalyGPT [35] Generates its own supervision by synthetically corrupting normal images with NSA/Poisson editing and training the decoder and prompt learner against auto-generated textual Q&A labels describing the injected anomalies.

5.2.3 Supervised (Weakly or Fully)

Supervised methods utilize labeled data to guide training, where annotations indicate the presence or classification of anomalies. In a fully supervised setting, labels are provided at the same granularity as the prediction target. In a weakly supervised setting, labels or a mixture of incomplete labels are provided at a coarser granularity than the prediction target, requiring the model to infer finer-grained predictions from broader supervision signals.

MIL [39] The model is trained under weak supervision, where video-level binary labels (normal/anomalous) guide a MIL ranking loss that learns to predict segment-level anomaly scores without temporal annotations.

5.3 Zero-Shot/ Training Free

Training free has no training loop involved for the core methodology. Technically these models usually contain a portion of the model that is pretrained on another training paradigm, but since the core methodology leverages this pretrained network and does no additional training of its own, it will fall under this categorization. Zero-shot inference extends this further, where the model is applied directly to a target domain or anomaly type it has never explicitly been exposed to, relying on the generalization capacity of the pretrained backbone.

PatchCore[26] PatchCore constructs its anomaly detector entirely at inference time by populating and compressing a memory bank of patch features extracted from a frozen pretrained backbone, requiring no task-specific training. PatchCore uses a frozen ImageNet-pretrained backbone with no training loop — the "training" phase consists entirely of extracting patch-level features from nominal images via forward passes and storing them in a coreset-subsampled memory bank, with anomaly scoring performed at inference through nearest-neighbor distance computation against this bank.

SPADE[27] SPADE is training-free, requiring only a single forward pass per training image to precompute and store frozen pretrained features, with all anomaly detection performed through nearest neighbor retrieval at inference time.

PaDiM[28] PaDiM is training-free, estimating Gaussian parameters (mean and covariance) per patch position through closed-form statistical computation over pretrained CNN embeddings of normal images, requiring no optimization or learning procedure.

PatchCoreCL[40] PatchCoreCL is training-free — it builds its anomaly detection capability through a single forward pass of normal images through a frozen pretrained backbone followed by coreset-based patch selection, with no optimization loop, loss function, or weight updates involved.

AnomalyDino[37] AnomalyDINO runs no training loop in its core methodology — a frozen, pretrained DINOv2 serves as the feature extractor, and the nominal memory bank is populated by a single forward pass over the reference images with no parameter updates, losses, or gradient steps involved.

6 Datasets

This section covers datasets that have been used as benchmarks or training data for anomaly detection methods. Each paper may have different interpretation of optimal use of the dataset and may only utilize a subset of the full set.

Table 3: Table demonstrates the relationship of usage between vision datasets and papers/methodologies that utilized the datasets
[1] Dataset created by the paper's authors or co-authors
[2] Papers that are not apart of the seminal works

Dataset	**Total**	**Type**	**Annotation**	**Category**	**Used By**
MVTecAD[43]	5,354 images	Image	Labels + pixel-level (1,888)	Industrial defects	[37] [26] [27] [28] [15] [29] [30] [16] [38][1] [33] [24] [25][2] [36][2] [35][2]
VisA[44]	10,821 images	Image	Labels + pixel-level	Industrial defects	[37] [25][2] [35][2]
MTD[45]	1,344 images	Image	Labels + pixel-level	Industrial defects	[26]
MNIST[46]	70,000 images	Image	Labels	General (digits)	[29] [16] [18] [19] [38] [32]
CUHK Avenue[47]	30,651 frames	Video	Bounding box	Surveillance (people)	[17] [18] [39] [13] [22] [23] [14] [24]
UCSD Peds 1 & 2[48]	98 video clips	Video	Frame-level labels + partial pixel masks	Surveillance (people)	[17] [18] [39] [13] [22] [23] [14] [24]
UBA[49]	230,275 patches	Image	Labels	Security (X-ray)	[19][1]
Woven Fabric[20]	250 images	Image	Labels	Industrial (textile)	[20][1]
SD-OCT[21]	290 volumes	Image (3D)	Labels	Medical (OCT)	[21][1]
UCF-Crime[39]	1,900 videos (128 hrs)	Video	Labels	Surveillance (crime)	[39][1]
Toy Pedestrian[13]	1,452 frames	Video	Labels	Synthetic (people)	[13][1]
Traffic[14]	5 scenes (∼200 min)	Video	Labels	Surveillance (traffic)	[14][1]
STC[50]	317,398 frames	Video	Frame-level labels	Surveillance (people)	[26] [27] [28] [17][1] [18] [13]
DTD[51]	5,640 images	Image	Labels	Textures	[15] [34]
ImageNet[52]	1,431,167 images	Image	Labels + bounding boxes	General	[15] [29] [38] [34]
DAGM[53]	16,100 images	Image	Labels	Industrial defects	[15]
CIFAR-10/100[54]	60,000 each	Image	Labels	General	[29] [30] [16] [31] [18] [19] [38] [32] [34] [33]
dSprites[55]	737,280 images	Image	Factor labels	Synthetic	[29]

Table 3: Table demonstrates the relationship of usage between vision datasets and papers/methodologies that utilized the datasets
[1] Dataset created by the paper's authors or co-authors
[2] Papers that are not apart of the seminal works

Dataset	**Total**	**Type**	**Annotation**	**Category**	**Used By**
BTAD[56]	2,540 images	Image	Labels + pixel-level	Industrial defects	[30]
FashionMNIST[57]	70,000 images	Image	Labels	Fashion	[16] [32] [33]
Caltech 256[58]	30,607 images	Image	Labels	General objects	[16] [34]
MovingMNIST[59]	200,000 frames	Video	None	Synthetic	[17]
NanoTWICE[60]	40 images	Image	Pixel-level	Industrial (SEM)	[20]
FFOB[61]	72,352 images	Image	Labels	Security (X-ray)	[19]
UMN[62]	∼6,450 frames	Video	Labels	Surveillance (crowd)	[39]
BOSS[63]	12 videos (∼48K frames)	Video	Labels	Surveillance (people)	[39]
ASIRRA[64]	25,000 images	Image	Labels	General (animals)	[32]
Subway[65]	186,650 frames	Video	Event-level labels	Surveillance (people)	[39] [22] [23]
Abnormal Crowd[66]	31 videos (∼43K frames)	Video	Labels	Surveillance (crowd)	[39]
SVHN[67]	99,289 images	Image	Labels	General (digits)	[34]
LSUN[68]	∼10M images	Image	Labels	Scenes	[34]
CUB-200[69]	11,788 images	Image	Labels + bounding boxes + parts	Fine-grained (birds)	[34] [33]
Stanford Dogs[70]	20,580 images	Image	Labels + bounding boxes	Fine-grained (animals)	[34]
Oxford-IIIT Pet[71]	7,390 images	Image	Labels + pixel-level + ROI	Fine-grained (animals)	[34]
Oxford 102 Flower[72]	8,189 images	Image	Labels + pixel-level	Fine-grained (flowers)	[34] [33]
Food-101[73]	101,000 images	Image	Labels	Fine-grained (food)	[34]

Table 3: Table demonstrates the relationship of usage between vision datasets and papers/methodologies that utilized the datasets
[1] Dataset created by the paper's authors or co-authors
[2] Papers that are not apart of the seminal works

Dataset	Total	Type	Annotation	Category	Used By
Places-365[74]	~1.84M images	Image	Labels	Scenes	[34]
TILDA[75]	3,200 images	Image	Labels	Industrial (textile)	[34]
WBC	484 images	Image	Labels	Medical (microscopy)	[33]
DIOR[76]	23,463 images	Image	Bounding boxes	Remote sensing	[33]
BMAD[77]	81,056 images/slices	Image	Labels + pixel-level (3 subsets)	Medical (multimodal)	[40][2]

6.1 Dataset Descriptions

MVTecAD[43] MVTec anomaly detection dataset contains 5354 high-resolution color images of ten objects and five textures with 73 different types of defects mimicking real-world industrial inspection scenarios: carpet, leather, bottles, pills, screws, transistors, etc. Dataset also contains pixel-accurate ground truth regions for 1888 images. This dataset may be used for classification and segmentation outputs. [43]

VisA[44] Comprised of 10,821 images total split into 9,621 normal and 1,200 anomalous samples with 12 different objects and anomaly subclassification. Classes include objects such as PCBs, candles, capsules, cashews, etc. [44]

MTD[45] Magnetic tile defects dataset contains a total of 1344 images with pixel level labeling for segmentation and defect classification such as crack, frays, break, uneven, etc. [45]

MNIST[46] Modified national institute of standards and technology database is a collection of handwritten digits from zero to nine, represented in 70,000 gray scales images that are 28 by 28 pixels. [46]

CUHK Avenue[47] Chinese University of Hong Kong avenue dataset is comprised of 16 training and 21 testing video clips captured at the Chinese University of Hong Kong campus avenue. 30652 total frames split into 15328 training and 15323 testing. Abnormal events are labeled with rectangular bounding boxes.[47]

UCSD Peds 1 & Peds 2[48] University of California San Diego pedestrian contains Peds 1 & Peds 2. Peds 1 contains 34 training and 36 testing video samples of groups of people walking towards and away from the camera at a top down view. Peds 2 contains 16 training and 12 testing video samples of pedestrian movement parallel to the camera plane. Each clip has corresponding ground truth annotations as an anomaly binary flag per frame. Pixel level binary masks are only present in a subset.[48]

UBA[49] University baggage anomaly dataset is a private dataset due to security reasons. This is a sliding window patched-based dataset comprised of 230,275 image patches derived from baggage X-ray imagery. 122,803 abnormal classes split among three classifications knife, gun, and gun components. [19]

Woven Fabric Texture Dataset[20] The dataset pertains to close up thread patterns of fabrics. The dataset contains 200 defect-free imagery split between two woven fabric texture types. With an additional 50 images containing various defects. Images are gray scaled of size 512 by 512 pixels. [20]

SD-OCT[21] This data set related to spectral-domain optical coherence tomography retinal volumes. There are a total of 290 clinical OCT volumes of resolution 496 by 512 by 49 voxel with 49 representing B-scans (image slices). 20 of the 290 are for testing with 50/50 split of healthy and pathological cases. [21]

UCF-Crime Dataset[39] This dataset includes 13 real world anomalies such as accidents, burglaries, explosions, fighting, etc in a total of 1900 videos equivalent to a runtime of 128 hours. This dataset was annotated by ten trained computer science vision experts of varying levels. The data was collected from Youtube and LiveLeak. [39]

Toy Pedestrian[13] A 2D oversimplified dataset of solid color stick figures and solid color vehicles on a white backdrop containing a total of 1452 frames split between 210 training frames and 1242 testing frames.[13]

Traffic Dataset[14] Collected from real-world traffic surveillance systems. Five different scenarios with each containing 30 minutes of normal and 10 minutes of testing footage. [14]

STC[50] Shanghai Tech campus dataset contains a total of 317,398 frames split into 274,515 training and 42,883 testing frames. There are 13 diverse scenes with 130 distinct abnormal events and approximately 5% of the total frames are abnormal frames. [50]

DTD[51] Describable textures dataset contains 5640 labeled images split among 47 categories depicting various textures like patterns, flowers, leaves, fabrics, etc. extracted from Google and Flickr. 120 images per category with image sizes from 300 by 300 to 640 by 640 pixels. Dataset contains a balanced split into train, validation, and test [51]

ImageNet[52] The most popular ImageNet subset contains a total of 1,431,167 images split into 1,281,167 training, 100,000 testing and 50,000 validation of general, generic, non-specific, images for the purposes of image classification and localization.[52]

DAGM[53] Deutsche Arbeitsgemeinschaft für Mustererkennung dataset contains ten classes of industrial surface defect high-resolution gray scale images with a mixture of defects and normal cases. There is a grand total of 16,100 images, but split between a development and competition dataset. 14,000 non-defective and 2,100 defective images. Each class is a 50/50 balanced split. Defective images contain exactly one defect per image. [78]

CIFAR-10, CIFAR-100[54] The Canadian Institute For Advanced Research datasets are labeled subsets of the 80 million tiny images dataset created by Krizhevsky, Nair, and Hinton. Both contain 60,000 32 by 32 color images split into 50,000 training and 10,000 testing samples. CIFAR-10 organizes images into 10 mutually exclusive classes such as airplanes, automobiles, birds, cats, and ships, while CIFAR-100 expands to 100 fine-grained classes further grouped into 20 coarse superclasses, providing both a fine and coarse label per image.[54]

dSprites[55] Disentanglement testing sprites dataset is a synthetically generated dataset designed to evaluate how well unsupervised models recover underlying generative factors. It contains 737,280 black and white 64 by 64 images procedurally generated from every possible combination of 6 independent latent factors: shape (square, ellipse, heart), scale, orientation, and x and y position. [55]

BTAD[56] BeanTech anomaly detection dataset is a real-world industrial anomaly dataset released by beanTech and the University of Udine, Italy. It contains 2,540 RGB images across three unspecified industrial products with varying resolutions of 1600 by 1600, 600 by 600, and 800 by 600 pixels respectively. The training set consists only of defect-free normal images, while the test set contains a mixture of normal and abnormal images, supporting both anomaly detection and pixel-level localization via ground truth masks. [30]

FashionMNIST[57] Fashion-MNIST is a dataset of grayscale apparel article images intended as a more challenging drop-in replacement for the original MNIST dataset. It contains 70,000 images split into 60,000 training and 10,000 testing examples across 10 clothing categories including t-shirts, trousers, pullovers, dresses, coats, sandals, shirts, sneakers, bags, and ankle boots. Each image is 28 by 28 pixels, matching the exact image size and train/test split structure of the original MNIST dataset. [57]

Caltech 256[58] Caltech-256 is a challenging object category dataset containing 30,607 images spread across 256 categories. It is an expansion of the original Caltech-101, with the number of categories more than doubled, a minimum of 80 images guaranteed per category, and the addition of a dedicated clutter category for testing background rejection. Images were collected from Google Images and manually screened for category fit. The dataset supports image classification and background rejection tasks. [58]

MovingMNIST[59] Moving MNIST is a synthetic video prediction dataset consisting of 10,000 sequences each of length 20 frames depicting two handwritten digits moving and bouncing within a 64 by 64 pixel grayscale frame. Intended as a standardized test set for sequence prediction and reconstruction tasks. [59]

NanoTWICE[60] NanoTWICE is a dataset of scanning electron microscope (SEM) images of electrospun nanofibrous materials intended for anomaly detection and defect localization. The dataset contains 40 test images of nanofibrous material surfaces, accompanied by a pre-trained sparse coding dictionary and anomaly detection density model. Defects manifest as structural irregularities in the fiber patterns. The dataset was introduced alongside a MATLAB GUI tool for pixel-level anomaly detection via sparse coding. [60]

FFOB[61] The Full Firearm vs. Operational Benign dataset is a UK government evaluation dataset comprising expertly concealed firearm threat items and operational benign non-threat imagery sourced from commercial X-ray security screening operations of baggage and parcels. The dataset contains 4,680 full firearm images as the abnormal class and 67,672 operational benign images as the normal class, with an 80/20 train/test split applied to normal samples. [19]

UMN[62] The University of Minnesota unusual crowd activity dataset consists of five different staged videos where people walk around and after some time start running in different directions. The anomaly present in the dataset is characterized solely by the running action. Each video averages 1,290 frames with a total dataset length of approximately five minutes. [39]

BOSS[63] The Behavior Observation for Safety and Security dataset was originally captured as part of the Eureka Celtic Initiative "On Board Wireless Secured Video Surveillance" project. It consists of video and audio recordings of acted actions inside a moving train using multiple calibrated cameras. The dataset contains 12 videos averaging 4,052 frames for a total runtime of approximately 27 minutes. Anomalies are performed by actors and include behaviors such as harassment, panic situations, fainting, fighting, and a person with a disease, alongside normal activities. It supports pose, action, and interaction recognition tasks. [63]

ASIRRA (Cats VS Dogs)[64] Animal Image Recognition for Restricting Access (ASIRRA) dataset originates from a CAPTCHA system originally deployed by Microsoft Research in partnership with Petfinder.com. The full image pool sourced from Petfinder.com consists of approximately three million color images of cats and dogs sourced from real animal shelter listings. The commonly used Kaggle subset contains 25,000 labeled images evenly split between 12,500 cats and 12,500 dogs for binary classification. Images vary in resolution but on average are 360 by 400 pixels, background, pose, and lighting, reflecting real-world photographic diversity rather than controlled capture conditions. [64]

Subway Entrance & Exit[65] Subway dataset contains two indoor surveillance video clips captured at a subway station divided into an Entrance and Exit sequence with a combined runtime of approximately 3 hours. The Entrance clip contains 121,749 frames and the Exit clip contains 64,901 frames. Anomalous events include pedestrians walking in the wrong direction and skipping payment. Annotations are provided as event-level labels rather than pixel-level masks and are evaluated on a per-event basis. [39]

Abnormal Crowd Dataset[66] Abnormal Crowd dataset contains 31 videos of crowded scenes with a combined runtime of approximately 24 minutes and an average of 1408 frames per video. Anomalous event categories include panic, fighting, congestion, obstacle, and neutral behaviors. [66]

SVHN[67] The Street View House Numbers dataset contains 99,289 real-world digit images cropped from Google Street View imagery, split into 73,257 training and 26,032 testing samples at 32 by 32 pixels in color, with an additional 531,131 less difficult extra training samples also available. Digits are labeled zero through nine and may appear alongside distracting digits in the periphery of each crop. [67]

LSUN[68] The Large Scale Scene Understanding dataset contains approximately 10 million labeled images across ten scene categories — bedroom, bridge, church outdoor, classroom, conference room, dining room, kitchen, living room, restaurant, and tower — and approximately 59 million labeled images across twenty object categories. Images are variable in resolution, distributed and released with the shorter dimension resized to 256 pixels. The dataset is intended for scene classification and object detection benchmarks. [68]

CUB-200[69] The Caltech-UCSD Birds 200-2011 dataset contains 11,788 images of 200 North American bird species split into 5,994 training and 5,794 testing images. Each image is annotated with bounding boxes, part locations, and attribute labels, making it a standard fine-grained visual recognition benchmark. [34]

Stanford Dogs[70] The Stanford Dogs dataset contains 20,580 images of 120 dog breeds sourced from ImageNet, split into 12,000 training and 8,580 testing images. The dataset is intended for fine-grained image categorization and features significant intra-class variation due to pose, lighting, and background diversity. [34]

Oxford-IIIT Pet[71] The Oxford-IIIT Pet dataset contains approximately 7,390 images across 37 cat and dog breed categories with roughly 200 images per class. Each image is annotated with a breed label, head region of interest (ROI), and pixel-level trimap segmentation mask, supporting both classification and segmentation tasks. [71]

Oxford 102 Flower[72] The Oxford 102 Category Flower dataset contains 8,189 images of flowers common to the United Kingdom split across 102 categories with between 40 and 258 images per class. Images exhibit large variation in scale, pose, and lighting and are accompanied by segmentation masks and class label annotations. [34]

Food-101[73] The Food-101 dataset contains 101,000 images evenly distributed across 101 food categories with 750 training and 250 manually verified testing images per class. Images were collected from foodspotting.com and intentionally include some amount of noise in the training split to reflect realistic web-scraped conditions. [34]

Places-365[74] The Places365 dataset is a large-scale scene-centric dataset containing approximately 1.8 million training images and 36,500 validation images across 365 scene categories such as airport, beach, forest, and shopping mall. Images are sourced from the broader Places database of over 10 million images and are intended for scene recognition and environmental context tasks. [34]

TILDA Textile Texture[75] The Textile Texture dataset (TILDA) is a reference dataset for evaluating visual inspection methods on textile surfaces. It contains 3,200 grayscale images of eight textile surface types at a resolution of 768 by 512 pixels. For each textile type, eight classes are defined — one defect-free reference class and seven defect classes — with 50 images per class. [75]

WBC The white blood cell dataset is a microscopy image collection comprised of 4 classes with 59 training and 62 testing images per class. The dataset depicts various white blood cell types at high magnification and is used to evaluate anomaly detection performance on out-of-domain medical imagery distinct from typical natural image datasets. [33]

DIOR[76] The dataset for object DetectIon in Optical Remote sensing images is a large-scale benchmark for object detection in optical remote sensing imagery. The dataset contains 23,463 images and 192,472 instances covering 20 object classes, and features a large range of object size variations in terms of both spatial resolution and inter- and intra-class size variability. Images are obtained under varying imaging conditions, weathers, seasons, and image quality, resulting in high inter-class similarity and intra-class diversity. [76]

BMAD[77] Benchmarks for Medical Anomaly Detection is a comprehensive evaluation benchmark consisting of six reorganized datasets from five medical imaging domains (brain MRI, liver CT, retinal OCT, chest X-ray, and digital histopathology). Contains a total of 81,056 images and slices sourced from BraTS2021, BTCV, LiTs, RESC, OCT2017, RSNA, and Camelyon16. Three datasets support pixel-level anomaly localization with segmentation masks, while the remaining three provide sample-level image labels. Benchmark also integrates fifteen state-of-the-art anomaly detection algorithms and three evaluation metrics (Image AUROC, Pixel AUROC, and PRO). [77]

6.2 Discussion

For popularity among the seminal works, it is clear that MVTecAD[43] has made a profound impact in the industrial manufacturing domain and may have been a contributing factor to the popularity of survey and research papers in that field. This conclusion can be applied for CUHK Avenue[47], UCSD Pedestrian[48], and STC[50] datasets. CIFAR (2009)[54], ImageNet (2009)[52], MNIST(1998)[46] being general datasets has seen a bit of usage considering their date of publication.

7 Discussion and Observations

This section sheds light on the broader challenges in anomaly detection, spanning the taxonomy of anomaly types, the impact of data quality and quantity on model development, the assumptions that underpin common detection approaches and their failure modes, and the various forms of model drift that arise when deployment conditions diverge from training conditions.

7.1 Anomaly Types

Anomalies may manifest across different aspects of image and video data, and can be broadly categorized by their occurrence pattern, extent, appearance, and semantic meaning. At the occurrence level, point anomalies represent single isolated events, contextual anomalies are only anomalous given their surrounding circumstances, and collective anomalies emerge from a group of instances that together form an irregular pattern[7]. Shifting to appearance, textural anomalies are characterized by surface-level deviations such as cracks, holes, and scratches, while functional anomalies impede function, such as missing or misaligned components.[1][2] Finally, at the extent level, local anomalies occupy only a small region of the image such as a lesion within healthy tissue, whereas global semantic anomalies span the image as a whole[10]. Together, these categories reflect that anomalies must be considered within context and at varying scales. More information about these types of anomalies are found under section 4.1.

7.2 Data Quantity and Quality

7.2.1 Dataset

Among the papers reviewed, we noticed a plethora of different datasets being used for evaluation and model construction. Upon further investigation, the contributing factor to the abundance of datasets with overlapping domains is the result of limitations within datasets motivating researchers to develop new datasets that mitigate those shortcomings. Limitations often deal with two super categories, quality and quantity. Generally speaking, more high-quality data that represents the intended target distribution is better. For instance insufficient variations in object types and unrealistic anomalies do not represent the real world which consequently encourages authors to develop more challenging datasets [39]. The sentiment for increasing diversity and variation is also echoed by others [50]. Moreover, authors may seek to pioneer a new search space by providing a high quality, large scale dataset that was previously absent [43]. From a different perspective, authors have introduced new dataset for their own use case to validate to performance and behavior of the model under controlled settings [13].

With regards to data quality, Cui et al. has mentioned that existing open datasets cover a limited number of scenarios which are not comprehensive enough and do not capture the nature of actual industrial diversity. Furthermore, the simplicity of data in datasets does not accurately reflect the complexities of industry [2]. This sentiment is mirror by Tao et al. who lists that some datasets are well imaged and uniformly illuminated which can inflate performance numbers as they do not mimic real complex industrial scenarios [1].

7.2.2 Data Influence on Models

Beyond general lack of data in terms of image count and video runtime [5][7], data limitations also deals with the nature of anomaly detection, which is the scarcity of anomalous data, especially for each researcher's domain of interest [2][11]. For industrial settings where "defects typically follow a long-tailed distribution"[3], this defect rarity issue is mitigated by leveraging the following techniques: pretrained networks, reconstruction, and data generation [3]. Pretrained networks which are trained on a generalized datasets include models such as Wide-ResNet, ImageNet, EfficientNet, etc. which theoretically results in the fundamental understanding of features like basic textures, shapes, and amalgamations forming distinct classes of everyday objects. This technique cuts down on the needs to train a model to learn from scratch thereby requiring less overall normal and abnormal data. However, it has been noted that this approach could negatively affect generalization abilities for different domains[2]. Reconstruction-based methods address the rarity problem by bypassing the need for anomaly data entirely. Rather than learning from anomalous examples, they learn the distribution of normal data, allowing anomalies to be identified by their failure to be reconstructed. Another approach is to augment anomalous data with fake data via data generation techniques. Anomalies generated through this method manifest as transformations to the image, cutting and pasting patches from and images and shuffling them around the image, or via natural synthetic anomalies.

7.3 Data and Model Assumptions

Every anomaly detection method relies on a set of assumptions about the underlying data distribution, and their performance can degrade sharply when those assumptions are violated. As an example, statistical-hypothesis-based methods for hyperspectral anomaly detection rest on the assumption that "background pixels follow a specific probability distribution, while anomalous pixels deviate from it."[4]

7.3.1 One Class

Perhaps the most pervasive assumption in the field is that anomaly detection is a one-class learning problem. Zaheer et al. articulated this assumption explicitly, noting that "anomaly detection problems are based on learning only normal data distribution and anything occurring outside that distribution is termed as anomaly. It can be seen as a one class problem, in which all other classes are unknown."[5] That notion is also shared by Duong et al.[8]. This framing is largely a consequence of practical necessity. As Zaheer et al. explain, "all the possible anomalies in a given environment cannot be predicted. Consequently, a training data based on abnormal examples cannot be generated. Hence, majority of the anomaly detection systems are based on the idea of learning a normal distribution against which test examples are matched to find anomalous events."[5] This assumption introduces a fundamental epistemic limitation: the model can only define anomalies in terms of deviation from learned normality, without any explicit understanding of what constitutes a meaningful anomaly versus a benign variation. This becomes particularly problematic in situations where small variances are acceptable, or cases where situational context is required to determine anomalous events.

7.3.2 Reconstruction

For instance, the usage of autoencoders and GANs in reconstruction methods operates under the intuitive assumption that learning on normal data will fail to reconstruct anomalous outputs as anomalous data is never learned. However, this theory has come under contention as Cui et al. states "reconstructors like autoencoders and GANs are highly generalizable and robust, and the anomalous part of the image can be well reconstructed, leading to hypothesis failure." [2]. Furthermore, this corroborated by Tao et al. who demonstrates that an anomalous scratch mark defect can actually be reconstructed in a slightly different form due to the over-generalization of the autoencoder. [1]. This issue occurs when the assumption about the data is violated. The data must contain a high degree of correlation and structure to be bottlenecked into a lower dimensional space[1]. Ideally the bottleneck preserves important features by compressing correlated patterns, while noise and low-salience features are discarded. An abundance of dissociated, overly complex features that do not correlate well require a larger latent space to map. Excessively increasing the bottleneck will cause the model to overfit and generalize to more features, which is the bias-variance trade off. Another perspective is to favor learning the template of normality and avoid learning visual primitives such as edges, textures and color gradients, which could contribute to an anomalous reconstruction. Research has continued in this area as Li et al.[3] highlights the work of Zhang et al. who proposed a DC-AE anomaly detection framework, "designed to address the issue of traditional autoencoders mistakenly reconstructing anomalies as normal features[3]." Generative adversarial approaches assume that the generator can sample data from the manifold of training data and thereby only reconstruct normal data from the manifold of normal data[2]. In practice, the assumption is fragile under low-data regimes. Tao et al. observed that CycleGAN-based methods suffer because "the lack of data samples poses a challenge in training two GAN networks well at the same time in industrial scenarios[1]."

7.3.3 Pretrained

Methods like PaDiM that utilized a pretrained network operate under a different set of assumptions. PaDiM[28] fits a multivariate Gaussian to per-location features; this formulation assumes that the dataset is spatially aligned such that the same pixel coordinate corresponds to the same semantic content across training images [1]. Assumptions such as this limit the deployment applicability in scenarios where the same object is positioned or posed differently. Normalizing flow methods assume that feature representations can be mapped via an invertible transformation to a known, tractable distribution, allowing exact likelihoods to be computed[1]. From this follows the anomaly detection assumption: normal samples get high likelihoods while anomalies get low ones. This works well when the flow is applied on top of a pretrained network, since the resulting features form a cleaner distribution that the flow can actually fit[1].

7.3.4 Neighborhood

Santhosh et al.[7] notes that distance-based approaches assume normal data occupy dense neighborhoods. This assumption that normal points are dense and anomalous points sparse breaks down in two opposite cases. The first is a false positive: legitimately normal points that reside in sparse regions of the distribution have few close neighbors and are misclassified as anomalies. The second is a false negative: anomalies that co-occur or cluster together become locally dense and fall below the anomaly threshold. When reframed, this same vulnerability extends to any method that defines normality through structural conformity whether by neighborhood, cluster membership, or learned manifold.

7.3.5 Visual Distinctness

Another assumption mentioned is visual distinction. In relation to VAEs, the assumption is that, "anomalies in images are always visually distinct from the background or normal regions. This may not always be the case, as certain types of anomalies may be visually similar to the background, making them harder to detect using this method."[2] This is not always the case as low contrast, visually harmonious anomalies can camouflage defects. In hyperspectral imaging, an anomaly may be spectrally distinct but spatially indistinguishable; in industrial inspection, a hairline crack may register as little more than a few pixels of slightly altered intensity.

7.4 Model Drift

7.4.1 Concept Drift

Santhosh et al.[7] observed that "normal behavior evolves over time." This means that data fit to a historical distribution may become gradually miscalibrated over time, this behavior is known as concept drift. Specifically defined as when the underlying mappings of input and output change typically over time, such that the same input may correspond to a different label than it did at training time. From another perspective, if an anomaly is defined as a rare occurrence, and that anomaly is frequently observed as occurring, then it no longer fits the definition of anomaly. The drift occurs when this happens over a period of time, otherwise the term shift may be more applicable.

The fundamental concept of continual learning addresses this issue directly, where the motivation is to enable life long learning capabilities while mitigating catastrophic forgetting of the prior distribution. Alternatively, the naive approach would be to retrain the network when the delta of drift becomes unacceptable. Taking PatchCoreCL[40] as an example, it maintains a per-task partitioned memory bank under a fixed global budget, re-applies coreset subsampling to preserve each task's feature coverage as allocations shrink, and uses a min-over-banks inference rule that performs implicit task identification at query time.

7.4.2 Covariate Drifts

Covariate or data drifts are finer-grained instances of domain drift. They are defined as the inputs to the network being different between training and test time while the model and real world distribution remains fixed. As an example, methods that assume a static scene or fixed object are highly susceptible to covariate drifts as slight changes in camera or object positions create new inputs that the model has never seen before. Another perspective, is that covariate shifts can serve as the anomaly signal for one class methods. If the model is only trained on normal samples, and a never seen before sample is introduced at inference time, it can induce a strong covariate shift (and drifts). The deviation can be used as the anomaly signal.

Methods that assume a stationary object or static scene may be subject to covariate (data) drift. This is caused by slight changes in camera angles, dirty lenses, or drastic scenery changes that do not match learned normal representation which will require recalibration via periodically retraining. Yu et al. frames these covariate drifts caused by fixed shutter speeds, light source intensity, image blur, etc. as interference factors[11]. Tao et al.[1] and Cui et al.[2] have partially shared similar sentiments.

An unfavorable instance of data drift is commonly addressed by multiple papers [4][2][7][11] and describes more of the situation of occurrence rather than a specific metric. The first issue is how benchmark and training conditions (typically normal conditions) do not match deployment, where datasets are too clinical and perfect which fails to represent actual production conditions. Models may exploit properties found in these benchmarks and training data thereby becoming overfit to the domain and failing to adapt to scenarios in real industrial environments[1]. This exploitation of certain features in the data can cause model performance variances from dataset to dataset. Second is the generalization argument: pretrained networks are biased toward their training data and do not generalize well to different domains[2].

7.4.3 Annotator Drifts

Label drifts can be caused during dataset creation, as Tao et al. mentions that "Annotation noise may be inadvertently introduced when labeling the data."[1] This is further exacerbated by the lack of discussion on labeling criteria and agreement discussions for some datasets. [9]. However, some datasets such as UCF-Crime Dataset[39], DTD[51], Places 365[74] and LSUN[68] have addressed annotator drift mitigation strategies with annotator training and some require that annotators to achieve a 90% or higher accuracy on a controlled set of images to be accepted. Datasets like BMAD[77], which is a collection of externally sourced medical images, inadvertently address label drifts by relying on labels inherited from their original source, which is typically certified medical professionals. Furthermore, web-sourced and generated imagery based on search queries inherit their labels programmatically from the query or generation code.

8 Future Direction and Consensus of Trend

Leveraging Anomaly Rarity Trends typically address the limitations of anomalies itself, particularly that anomalies are rare. Some methods like one-class classification, reconstruction, memory banks, etc. leverage this rarity when sampling from the distribution where a low probability density region equates to the flagging of anomaly. When viewed under a clustering perspective, anomalies are typically present in low-density neighborhoods which are distant from the well-formed clusters. In principle, this is exploiting covariate shifts to generate an anomaly signal. However, issues arise when the clusters and distribution are malformed due to misguided or overly simplistic training data[1][2]. As mentioned in section 7.2, the quantity and quality of a dataset is paramount to achieving a realistic evaluation of a model's performance. While shared datasets are useful for comparing various approaches, idealized datasets, which do not mimic the variability of real world conditions, can overinflate model scores without reflecting the domain drift that would appear in production. Therefore, it is recommended to use a dataset that mimics the exact deployment scenarios with similar variability in features such as lighting conditions.

Data Augmentation Moreover, data issues are further addressed via technical approaches such as data augmentations [11][2]. Approaches involve augmentation via rearranging patches of an image on a fixed grid, rotations, or synthetic textures generated from Perlin noise. The introduction of high quality realistic anomalous data by image generation models could induce a paradigm shift in image anomaly detection if proven to be highly effective. While some authors claim it is intractable to realistically cover all cases[5][8], improvements in synthetic image generation such as image inpainting models may be a future direction to bridge the gap. Furthermore, generating known anomalies for domain specific use cases [11] where the classification of distinct anomaly types can be easily done may prove useful.

Foundation Models and Multimodal Approaches Another method that provides additional signal without artificial data generation involves foundation models and the introduction of additional modalities. Foundation models, classified as representation-based in our taxonomy, are pretrained on large general datasets and have proven tremendously popular, 15 papers in this review utilize the approach. Their broad generalization capabilities assist in downstream tasks and reduce the burden of collecting large amounts of domain-specific training data. Cui et al.[2], [3], and Cai et al.[10], each having surveyed the landscape, believe more research in this direction is warranted, including into pretrained vision-language models. VLMs in particular offer a dual advantage: pretrained weights that can be fine-tuned for specific tasks, and inherent multimodality that enriches feature representations. More broadly, the introduction of additional modalities beyond VLMs is also a promising avenue for improving detection performance [1] [42].

Under Explored Areas Advancing discovery should not only be limited to model techniques, but domains as well. Based on visual anomaly detection survey papers, there seems to be a heavy emphasis on societal monitoring, inclusive of public safety, people and roads. Second most popular being manufacturing. The most popular type of dataset used by papers is MVTecAD (Industrial) and pedestrian walking (Social) datasets, which is to be expected as datasets were derived from seminal works which were derived from the survey papers. Techniques used in these fields have potential to be cross compatible with adjacent domains. Take for instance, social monitoring typically deals with a static background while foreground pedestrians or vehicles transverse across the scene. Abstractly this is akin to natural disaster monitoring and result in similar challenges such as lens degradation via dirt or lighting variations. Utilizing techniques from explored tasks and domains gives an opportunity to help develop non-predominant fields of study.

Conclusion Based on each author's perspective, pertinent to their domain and observations, there is no unanimous convergence of future directions. However, there are general patterns and consensus for which a subset of authors agree upon. The common applied approach is from the curation of the dataset with regards to quality, in practice, the dataset should have a similar distribution to its deployment application. For building a pipeline, the proven approach is to utilize representation based approaches which can be coupled with a pyramid for varying spatial resolutions. This can be fed into a one class classifier or into a reconstruction method while being cautious of the assumptions. Alternatively, one may pursue the state of the art and bleeding edge, vanguarding experimentation with VLM and multi-modalities, which is a potential direction for progress.

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